

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
	(f)		Nitrogen boils OR heat transferred to LN ₂ OR high pressure inside bottle. Accept gas expands	1			1		
	(g)		Force on bottle / nozzle is equal and opposite to force on gas / nitrogen (1) Provides [anticlockwise] moment / torque [for rotation] (1) For 2 nd mark, do not accept “principle of moments” because this is relevant to equilibrium or adding moments. Don't accept equal and opposite forces acting on the walls	1	1		2		
	(h)		Sound [waves] produced (or implied) (1) Implication of pressure release / molecule or gas release / energy release OR displacement of particles/molecules OR vibration of particles / molecules (1)	1	1		2		
	(i)		Nitrogen escapes from the container OR nitrogen escapes into lift (1) ACCEPT pressure in lift builds up DON'T ACCEPT container explodes OR container acts like rocket Leading to suffocation / anything to do with breathing N ₂ (1)			2	2		
			Question 6 total	3	5	12	20	3	0